

Shouvik Mondal

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ABOUT ME

I am a PhD scholar in Computer Science at the Indian Statistical Institute (ISI), Kolkata, with seven years of mathematical training at ISI Bangalore through the B.Math, M.Math, and MS QMS programs. My mathematical training has included Combinatorics, Number Theory, Real Analysis, Abstract Algebra, Geometry, and Graph Theory. I particularly enjoy solving and discussing challenging mathematical problems, and have experience mentoring students preparing for Mathematics Olympiads and competitive entrance examinations.

MATHEMATICAL INTERESTS

Mathematics Number Theory, Combinatorics, Algebra, Calculus, Real Analysis, Geometry, and Graph Theory.

Research Computational Geometry, and Parametrized, Exact exponential-time, and Approximation Algorithms, with a focus on geometric and graph theoretic problems.

EDUCATION

Indian Statistical Institute, Kolkata

Jul 2025 – Present

PhD in Computer Science

India

- Junior Research Fellow (JRF), affiliated with the Advanced Computing and Microelectronics Unit (ACMU), under the supervision of Prof. Sasanka Roy.
- Coursework includes Computer Organization, Data Structures and Algorithms, Theory of Computation, Design and Analysis of Algorithms, and Advanced Algorithms and Computational Topology.

Indian Statistical Institute, Bangalore & Hyderabad

Jul 2022 – May 2024

MS in Quality Management Science (MS QMS)

India

- 94.90% – First Division with Distinction.
- Four-semester program in Quality Management and its applications, with an emphasis on Quantitative Methods and Applied Statistics.

Indian Statistical Institute, Bangalore

Jul 2019 – May 2021

Master of Mathematics (M.Math)

India

- 83.80% – First Division with Distinction.
- Coursework included Abstract Algebra, Analysis, Measure Theory, Topology, Geometry, Number Theory, and Combinatorics.
- Studied Operator Theory under Professor Jaydeb Sarkar as an elective project.

Indian Statistical Institute, Bangalore

Jul 2016 – May 2019

Bachelor of Mathematics (B.Math)(Hons.)

India

- 75.00% – First Division with Distinction.
- Coursework included Abstract Algebra, Analysis, Topology, Geometry, Number Theory, and Graph Theory, alongside Physics, Statistics, Computer Science, and Probability.

Bidhannagar Government High School, WBCHSE

Apr 2016

Higher Secondary (Class XII)

West Bengal, India

- 95.20% in Science stream (PCM), with Statistics as an elective.

RESEARCH & PUBLICATIONS

Perfectly Guarding Straits: Exact Algorithms for Weak Visibility Polygons

Shouvik Mondal, *Udvas Das, and Sasanka Roy*

2026

- arXiv preprint, [arXiv:2609.15510](#) [cs.CG].

Witness Set in Weak Visibility Polygons is Polynomial-Time Solvable

Udvas Das, Shouvik Mondal, and Sasanka Roy

2026

- arXiv preprint, [arXiv:2609.24460](#) [cs.CG].

Enhancing Supervised Learning Through Design of Experiments for Optimal Missing Value Imputation

Shouvik Mondal and *Boby John*

2024

- Published as a technical report at SQC & OR Unit, Indian Statistical Institute, Kolkata.

TEACHING & MENTORSHIP EXPERIENCE

Olympiad and Entrance Examination Mentor

2025 – Present

Independent

India

Mentored students preparing for Mathematics Olympiads and entrance examinations for the Indian Statistical Institute (ISI) and Chennai Mathematical Institute (CMI).

Subject Matter Expert, Advanced Mathematics

Oct 2018 – Apr 2023

Chegg India Pvt Ltd

India

Served as a Subject Matter Expert in Advanced Mathematics, solving and explaining mathematics problems for students worldwide across a wide range of topics.

PROFESSIONAL EXPERIENCE

PricewaterhouseCoopers (PwC) LLP India

Jul 2024 – Oct 2025

Associate, Technology Consulting

Kolkata, India

- Worked with industry clients on data analysis and practical problem-solving projects.
- Used statistical and computational methods to study real-world data and communicate findings clearly to clients.

Louis Dreyfus Company (LDC) India

Jan 2024 – Jun 2024

Data Science Intern

India

- Worked on statistical modeling and data analysis related to agricultural yield prediction and resource optimization.

AWARDS & ACADEMIC ACHIEVEMENTS

1. CSIR UGC-JRF (Junior Research Fellowship) awardee – All India Rank (AIR) 81 in Mathematical Sciences, 2021.
2. All India Rank (AIR) 3 in the Indian Statistical Institute entrance examination for MS in Quality Management Science, 2022.
3. All India Rank (AIR) 5 in the Indian Statistical Institute entrance examination for PhD in Computer Science, 2025.